

$CM = 181,44 \text{ Tn}$ $CV = 105,23 \text{ Tn}$ $F'c = 3512 \text{ kg/cm}^2$ $F_y = 4200 \text{ kg/cm}^2$

→ Cuantía de Acero 5% Diseñar la columna cuadrada

$P_g = \frac{A_{st}}{A_g}$ $A_{st} = P_g A_g$ $P_g = 5/100 = 0,05$

$q_u = 1,4(181,44) + 1,7(105,23)$
 $q_u = 432,907 \text{ tn}$
 $= 0,7(0,80)$

Loza aligerada
 Lozas Manjar
 Viga reforzada
 repuestas
 Columnas cuadradas
 Axialmente

CONCRETO ARMADO II

Del segundo Ejemplo $P_u = 179,98 \text{ toneladas}$

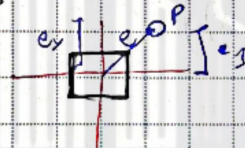
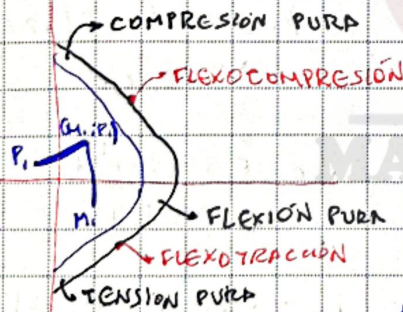
Ejemplo 3 $\phi = 0,85$ $P_g = 2\%$

UNIDAD 1: DISEÑO DE LOZA HACIZA EN DOS DIRECCIONES

→ Método DE COEFICIENTES ACI → MÉTODO DE PÓRTICO EQUIVALENTE

UNIDAD 2: COMPORTAMIENTO Y DISEÑO DE COLUMNAS CON CARGA BIAJIAL Y DISEÑO DE CIMIENTOS Y ZAPATAS AISLADAS

SE PRESENTA TRABAJO



UNIDAD 3: COMPORTAMIENTO Y DISEÑO DE ZAPATAS COMBINADAS, ZAPATAS CONECTADAS Y MUROS DE CONCRETO ARMADO

· VIGA DE REDISTRIBUCIÓN DE MOMENTOS
 · DISEÑO DE PLACAS

(Libros) Teodoro Hansen, Otazzi Pajini

DISEÑO EN C° A° → SECCIÓN RECTANGULAR

DISEÑO SIMPLEMENTE REFORZADA ① CASO I

DISEÑO DOBLEMENTE REFORZADA ② CASO II

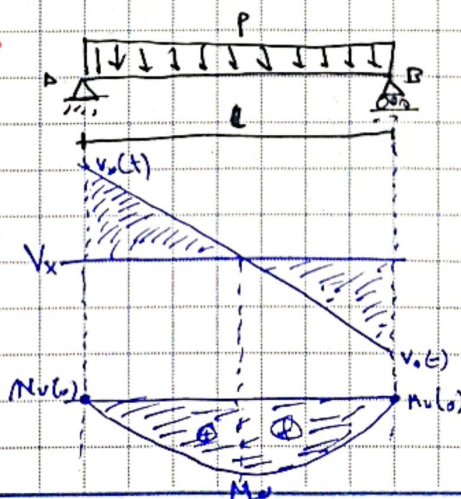
① $M_u \leq M_{ur \text{ max}}$

EN FUNCIÓN DEL ANÁLISIS ESTRUCTURAL

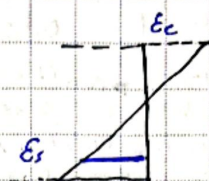
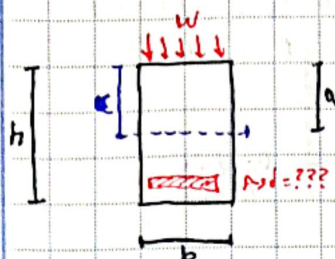
EN FUNCIÓN DEL DISEÑO ESTRUCTURAL

① h A_{sd}, P

② h A_{sd}, P A_{sd}, P



CASO I:



Test: porque no se dueña en el periodo de diseño

Test: encontrar el modelo matemático $\beta_1 = 0.85$ $C = 0.85 f_c$

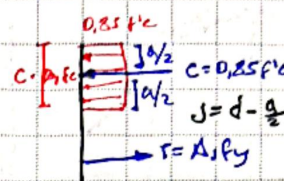


DIAGRAMA DEFORMACIONES

ALTURA BLOQUE COMPRESIONES

$$M = Fx d$$

$$M = T x d$$

$$M_m = A_s f_y \left(d - \frac{a}{2} \right) \text{ MOMENTO NOMINAL}$$

$$\phi M_m = M_u R = \phi A_s f_y \left(d - \frac{a}{2} \right) \text{ MOMENTO RESISTENTE}$$

COMPRESION = TRACCION

$$0.85 f_c (a) b = A_s f_y$$

$$a = \frac{A_s f_y}{0.85 f_c (b)}$$

$$A_s = f_b d$$

$$a = \beta_1 c$$

$$c = \frac{a}{\beta_1} \Rightarrow \text{ALTURA EN EJE NEUTRO}$$

CANTIA = % QUE NO PUEDE SUPERAR

$$\beta_1 = 0.85 \rightarrow f_c = 2800 \text{ kg/cm}^2$$

$$\beta_1 = 0.80 \rightarrow f_c = 3500 \text{ kg/cm}^2$$

$$\beta_1 = 0.75 \rightarrow f_c = 4200 \text{ kg/cm}^2$$

1) PARAMETRO MINIMO

$$P_{min} \rightarrow A_{smin} \rightarrow a_{min} \rightarrow C_{min} \rightarrow M_{URmin}$$

$$E.060 \rightarrow P_{min} = \frac{0.70 (f_c)}{f_y} \rightarrow A_{smin} = P_{min} b d \rightarrow a_{min} = \frac{A_{smin} f_y}{0.85 f_c b} \rightarrow C_{min} = \frac{a_{min}}{\beta_1} \rightarrow M_{URmin} = \phi A_{smin} f_y \left(d - \frac{a_{min}}{2} \right)$$

$$\rightarrow M_{URmin} = \phi A_{smin} f_y \left(d - \frac{a_{min}}{2} \right)$$

$$\rightarrow A.C.I \Rightarrow P_{min} = 14 \text{ fy}$$

2) PARAMETRO BALANCEADO

$$P_b = 0.85 \beta_1 \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \rightarrow A_{sb} = P_b (b) d \rightarrow a_b = \frac{A_{sb} \cdot f_y}{0.85 \cdot f_c \cdot b} \rightarrow \beta_1 = \frac{a_b}{\beta_1} \rightarrow M_{URb} = \phi A_{sb} f_y \left(d - \frac{a_b}{2} \right)$$

3) PARAMETRO MAXIMO

$$P_{max} = K P_b \left\{ \begin{array}{l} K=0.75 \rightarrow C-9 \Rightarrow A_{smax} = P_{max} \cdot b \cdot d \rightarrow a_{max} = \frac{A_{smax} f_y}{0.85 (f_c) b} \rightarrow C_{max} = \frac{a_{max}}{\beta_1} \Rightarrow M_{URmax} = \dots \\ K=0.50 \rightarrow SISMO \end{array} \right.$$

$$\dots M_{URmax} = \phi A_{smax} f_y \left(d - \frac{a_{max}}{2} \right) \text{ PROBLEMA DE APLICACION #01}$$

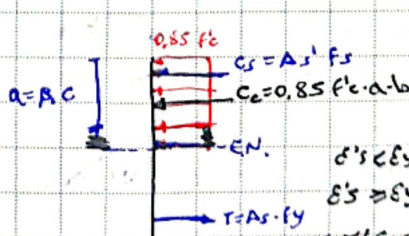
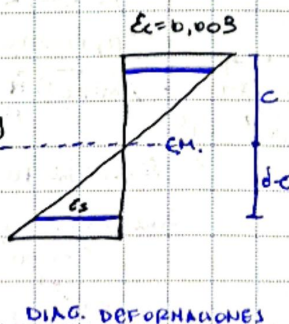
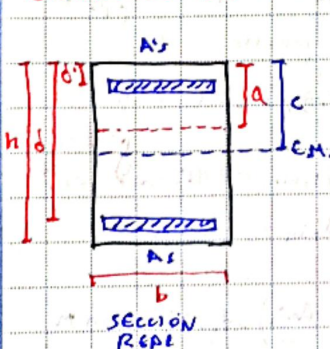
EN EL SIGUIENTE MODELO ESTRUCTURAL, DISEÑAR LA VIGA DE SECCION RECTANGULAR DE CONCRETO ARMADO, MOSTRANDO EN EL ADJUNTO, CUYAS CARACTERISTICAS GEOMETRICAS Y TECNICAS SON:

$$b = 30 \text{ cm}, h = 60 \text{ cm} \quad f'_c = 210 \text{ kg/cm}^2, f_y = 4200 \text{ kg/cm}^2$$

NOTA: PARA EL DISEÑO USE LA N.T.P. E-060 DEL RNE PARA CARGAS DE GRAVEDAD

NO VALE

CASO II: ANALISIS Y DISEÑO DE SECCIONES DOBLEMENTE REFORZADAS, SOMETIDO A ESFUERZOS DE FLEXION

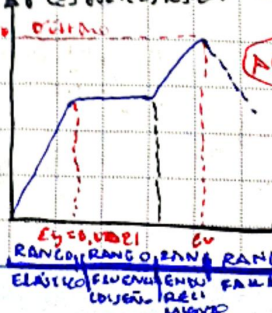


$$f_y = \epsilon_s \cdot E_s$$

$\epsilon'_s < \epsilon_y \rightarrow A_s' \text{ No Fluye}$
 $\epsilon'_s \geq \epsilon_y \rightarrow A_s' \text{ Fluye}$

$f'_s \leq f_y \rightarrow A_s' \text{ No Fluye}$
 $f'_s = f_y \rightarrow A_s' \text{ Fluye}$

DIAGRAMA DE FUERZAS



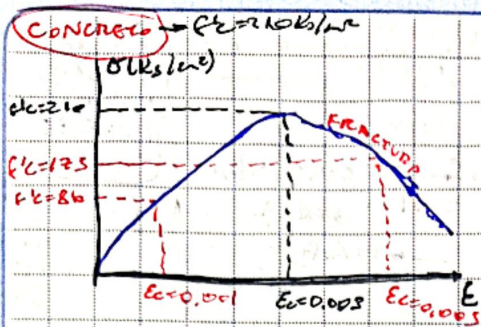
DEFORMACION DEL ACERO

$$f_y = \epsilon_y E_s$$

$$\epsilon_y = \frac{f_y}{E_s} \rightarrow \frac{4200}{21000} \rightarrow \epsilon_y = 0.0021$$

DEF. UNITARIA

Deformación



Asumir $A's$ y A_s ESTAN EN FLUJENIA $\Rightarrow \epsilon'_s \geq \epsilon_y \Rightarrow f'_s = f_y$
Por condición de equilibrio $\Rightarrow$ COMPRESION = TRACCION

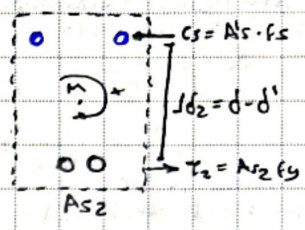
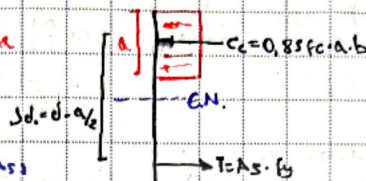
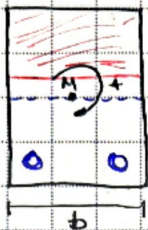
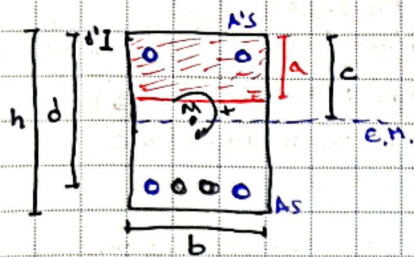
$$C_c + C_s = T \quad 0.85 \cdot f'_c \cdot a \cdot b + A's \cdot f'_s = A_s \cdot f_y$$

$$Q = \frac{A_s \cdot f_y - A's \cdot f'_s}{0.85 \cdot f'_c \cdot b} \Rightarrow Q = \frac{(A_s - A's) \cdot f_y}{0.85 \cdot f'_c \cdot b}$$

Asumir ACERO NO FLUYE $\epsilon'_s < \epsilon_y \Rightarrow f'_s \neq f_y$

$$C_c + C_s = T \quad 0.85 \cdot f'_c \cdot a \cdot b + A's \cdot f'_s = A_s \cdot f_y$$

$$Q = \frac{A_s \cdot f_y - A's \cdot f'_s}{0.85 \cdot f'_c \cdot b}$$



$A's \neq A_s \Rightarrow A's \rightarrow$ No fluye

$$f'_s < f_y$$

$$\epsilon'_s < \epsilon_y$$

$A's = A_s \Rightarrow A's \rightarrow$ Fluye

$$f'_s = f_y$$

$$\epsilon'_s \geq \epsilon_y$$

$$M_{n1} = T \cdot Jd_1$$

$$M_{n1} = A_s \cdot f_y \cdot (d - \frac{a}{2})$$

$$M_n = M_{n1} + M_{n2}$$

$$M_{n2} = C_s \cdot Jd_2$$

$$M_{n2} = A's \cdot f'_s \cdot (d - \delta')$$

$$M_n = A_s \cdot f_y \cdot (d - \frac{a}{2}) + A's \cdot f'_s \cdot (d - \delta')$$

$$\phi M_n = M_{UR}$$

$$M_{UR} = \phi A_s \cdot f_y \cdot (d - \frac{a}{2}) + \phi A's \cdot f'_s \cdot (d - \delta')$$

Momento Nominal Real

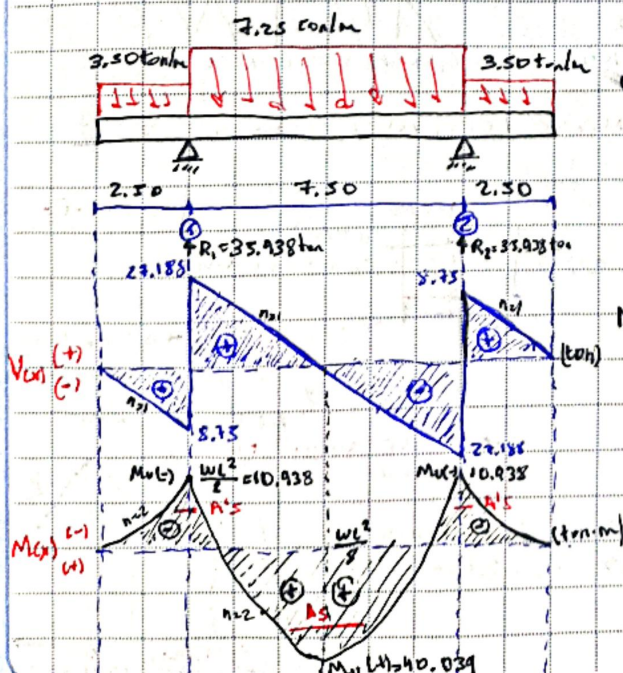
Momento último resistente Normativo

PROBLEMA DE APLICACIÓN N° 01

EN EL SIGUIENTE MODELO ESTRUCTURAL DISEÑAR LA VIGA DE SECCIÓN RECTANGULAR DE CONCRETO ARMADO MOSTRADO EN EL DETALLE ADJUNTO, CUYAS CARACTERÍSTICAS SON:

$$b = 30 \text{ cm} \quad h = 60 \text{ cm} \quad f'_c = 210 \text{ kg/cm}^2 \quad f_y = 4200 \text{ kg/cm}^2$$

NOTA: USE N.T.P. E. 060. PARA CARGAS DE GRAVEDAD



DATOS DE DUEÑO
 $M_U(L) = 10.938 \text{ ton-m}$ $f'_c = 210 \text{ kg/cm}^2$
 $M_U(H) = 40.039 \text{ ton-m}$ $f_y = 4200 \text{ kg/cm}^2$
 $b = 30 \text{ cm}$
 $h = 60 \text{ cm}$
 $\rho = 0.85, K = 0.75$
 $d = \begin{cases} 1 \text{ capas} \rightarrow d_1 = h - 6 = 54 \text{ cm} \\ 2 \text{ capas} \rightarrow d_2 = h - 9 = 51 \text{ cm} \end{cases}$

1) PARAMETRO MINIMO

$$\rho_{min} = \frac{0.70 \sqrt{f'_c}}{f_y} \Rightarrow \rho_{min} = 0.00242$$

$$A_{smin} = \rho_{min} \cdot b \cdot d \Rightarrow A_{smin} = 9.912 \text{ cm}^2$$

$$A_{min} = \frac{A_{smin} \cdot f_y}{0.85 \cdot f'_c \cdot b} \Rightarrow A_{min} = 3.068 \text{ cm}$$

$$M_{URmin} = \phi A_{smin} \cdot f_y \cdot (d - \frac{A_{min}}{2}) = 7.75961 \text{ ton-m}$$

06 Abril → Presentar primer Avance
11 Mayo → 5to Avance

08 de Junio → Tercer Avance
08 de Junio → Sustentación

MARK BOOK

② PARÁMETRO MÁXIMO (CAPACIDAD MÁX. RESISTENCIA)

$$P_b = 0.85 \rho \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow 0.02125$$

$$P_{máx} = R \cdot P_b = 0.75 (0.02125) \Rightarrow P_{máx} = 0.01594$$

$$A_{smáx} = P_{máx} \cdot b \cdot d \Rightarrow A_{smáx} = 24.384 \text{ cm}^2 = A_{sg}$$

$$\Delta_{smáx} = \frac{A_{smáx} \cdot f_y}{0.85 \cdot f_c \cdot b} \Rightarrow \Delta_{smáx} = 19.125 \text{ cm}$$

$$C_{máx} = 22.50 \text{ cm} = C$$

$$M_{Rmáx} = \phi A_{smáx} f_y \left(d - \frac{\Delta_{smáx}}{2} \right) = 38.1946 \text{ ton} \cdot \text{m} = M_u$$

$$M_u \leq M_{Rmáx} \rightarrow \text{D.R.} \quad \begin{cases} M_u = 10.938 \text{ ton} \cdot \text{m} & \text{D.R.} \\ M_u = 40.039 \text{ ton} \cdot \text{m} & \text{D.D.R.} \end{cases}$$

$$M_u > M_{Rmáx} \rightarrow \text{D.D.R.}$$

③ DISEÑO APOYO 1 y 2 $M_u = 10.938 \text{ ton} \cdot \text{m}$

$$① 0.59 w^2 - w + \frac{M_u \cdot (1)}{\rho_b \cdot d^2 \cdot f_c} = 0 \Rightarrow w = 0.06896$$

$$② \rho_b = w \frac{f_c}{f_s} = 0.00345$$

$$③ A_{sd} = \rho_b \cdot b \cdot d = 3.586 \text{ cm}^2$$

④ DISEÑO TRAMO 1-2

DATOS: $M_{u1} = 40.039 \text{ ton} \cdot \text{m}$
 $f_c = 240 \text{ kg/cm}^2$
 $f_y = 4200 \text{ kg/cm}^2$
 $b = 30 \text{ cm}$ $d = 51 \text{ cm}$ $d - b = 21 \text{ cm}$

$$M_u = M_{u1} + M_{u2}$$

$$40.039 = 38.1946 + M_{u2} \Rightarrow M_{u2} = 1.8444 \text{ ton} \cdot \text{m}$$

$$M_{ur} = \phi A_{s1} f_s \left(d - \frac{\Delta}{2} \right) + \phi A_{s2} \cdot f_s (d - d')$$

SECCIÓN REAL M_{u1} SECCIÓN IMAGINARIA M_{u2}

ASUMIR
A'S FLUYE
 $f_s = f_y$

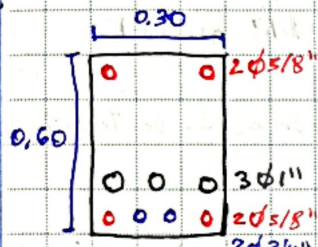
$$M_{u2} = \phi A_{s2} \cdot f_s (d - d') \Rightarrow A_{s2} = \frac{M_{u2}}{\phi f_s (d - d')} = \frac{1.089 \text{ cm}^2}{1} \Rightarrow A_{s2} = 1.089 \text{ cm}^2$$

$$E_c = 0.003 \quad \frac{\epsilon_c}{C} = \frac{\epsilon_s}{C - d'}$$

$$\epsilon_s = \frac{\epsilon_c (C - d')}{C} = \frac{0.003 (22.50 - 6)}{22.50} = 0.0022$$

$$\text{Si } \epsilon_s = 0.0022 > \epsilon_y = 0.0021 \Rightarrow A's \text{ fluye}$$

$$A_s = A_{s1} + A_{s2} = 24.384 + 1.089 \Rightarrow A_s = 25.468 \text{ cm}^2$$



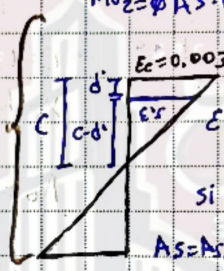
$$A_s = 25.468 \text{ cm}^2$$

$$A_{smin} = 4.00 \text{ cm}^2$$

$$A_{sr} = 21.468 \text{ cm}^2$$

$$3 \phi 1" = 3 \times 5.10$$

$$2 \phi 3/4" = 2 \times 2.84$$



EN CASO "NO FLUYE"

$$A_{s2} \cdot f_s = A_{s2} \cdot f_y$$

$$A_{s2} = A_{s2} \cdot \frac{f_y}{f_s}$$

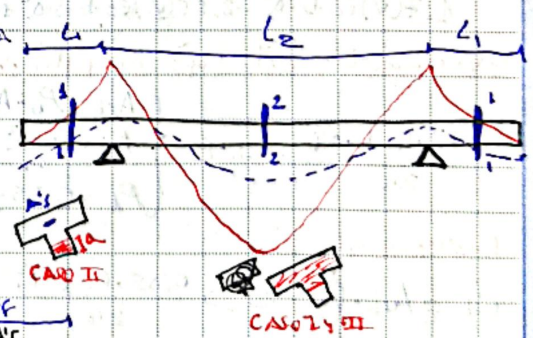
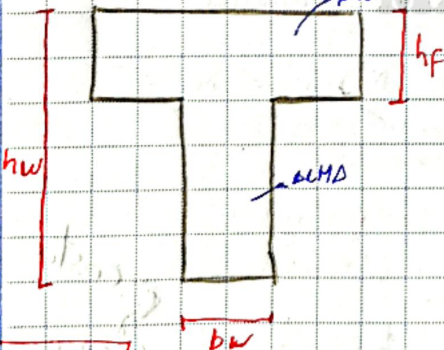
$$f_s = \epsilon_s \cdot E_s$$

ANÁLISIS Y DISEÑO DE SECCIONES TIPO "T" Y "L" SOMETIDOS A ESFUERZOS DE FLEXIÓN

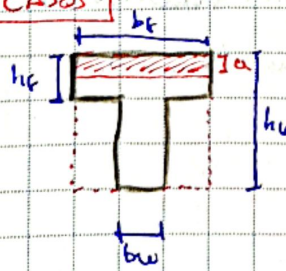
b_f = base de ala
 b_w = base del alma
 h_f = altura de ala
 h_w = altura de alma

CONTINUA

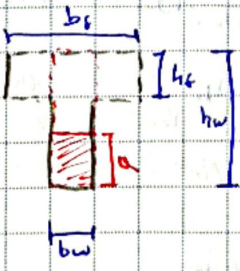
AISSADA



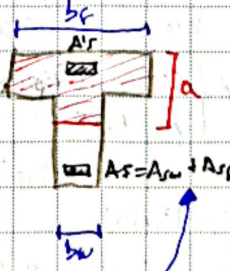
CASOS



CASO I
 $a \leq h_f$



CASO II
 $b_{f1} = b_w$

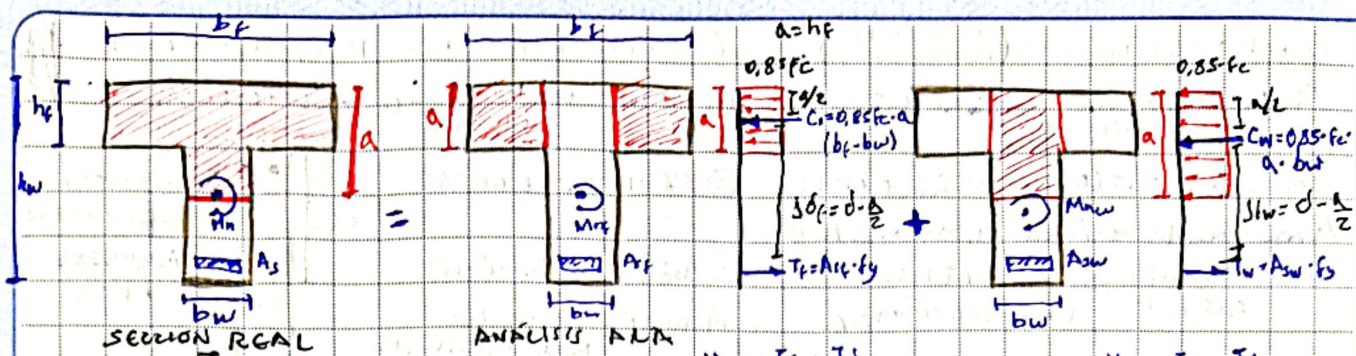


$$A_s = A_{sw} + A_{sf}$$

$$M_u = M_{uf} + M_{uw}$$

DISEÑO DE SECCIONES RECTANGULARES D.R. y D.D.R.

ANÁLISIS Y DISEÑO COMO SECCIÓN T y L



$$M_n = M_{nf} + M_{nw} = A_{sf} \cdot f_y \left(d - \frac{h_f}{2}\right) + A_{sw} \cdot f_y \left(d - \frac{a}{2}\right)$$

$$A_s = A_{sf} + A_{sw}$$

$$M_{nf} = T_f \cdot J_d$$

$$M_{nf} = A_{sf} \cdot f_y \left(d - \frac{h_f}{2}\right)$$

$$C_f = T_f$$

$$0.85 \cdot f_c \cdot a \cdot (b_f - b_w) = A_{sf} \cdot f_y$$

$$A_{sf} = \frac{0.85 \cdot f_c \cdot h_f (b_f - b_w)}{f_y}$$

$$M_{nw} = T_w \cdot J_d$$

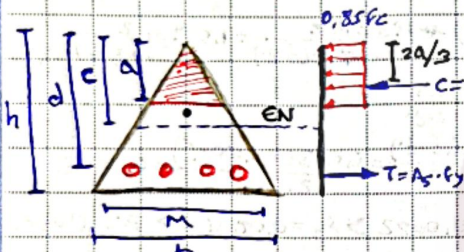
$$M_{nw} = A_{sw} \cdot f_y \left(d - \frac{a}{2}\right)$$

$$C_w = T_w$$

$$0.85 \cdot f_c \cdot a \cdot b_w = A_{sw} \cdot f_y$$

$$A_{sw} = \frac{0.85 \cdot f_c \cdot a \cdot b_w}{f_y}$$

VIGAS CON SECCIONES DIFERENTES



$$c = \frac{0.85 \cdot f_c \cdot b \cdot a^2}{2h}$$

$$T = A_s \cdot f_y$$

Por equilibrio: $C = T$

$$\frac{0.85 \cdot f_c \cdot b \cdot a^2}{2h} = A_s \cdot f_y \Rightarrow A_s = \frac{0.85 \cdot f_c \cdot b \cdot a^2}{2 \cdot f_y \cdot h} \quad (1)$$

MOMENTO DE RESISTENCIA: M_n

$$M_u = \phi \cdot \frac{0.85 \cdot f_c \cdot b \cdot a^2}{2h} \left(d - \frac{2a}{3}\right); a = \beta_1 \cdot c \quad (2)$$

Debe calcularse " ϕ " para secciones triangulares

$$1.7(\phi) f_c \cdot b \cdot a^3 - 2.55(\phi) f_c \cdot b \cdot d \cdot a^2 + 6h \cdot M_u = 0 \quad (3)$$

Verificación del tipo de falla en estado balanceado: $C_b = T_b$

$$\frac{0.85 \cdot f_c \cdot b \cdot a_b^2}{2h} = A_{sb} \cdot f_y \quad \begin{cases} A_{sb} = P_b \cdot A_d \\ A_d = \frac{b \cdot d^2}{2h} \\ P_b = 0.85 \cdot \beta_1 \cdot \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y}\right) = \frac{\beta_1 \cdot f_c}{2 f_y} \end{cases}$$

Reemplazando:

$$C_b = A_{sb} \cdot f_y \Rightarrow \frac{0.85 \cdot f_c \cdot b \cdot a_b^2}{2h} = \frac{\beta_1 \cdot f_c}{2 f_y} \left(\frac{b \cdot d^2}{2h}\right) f_y$$

$$\text{Reduciendo: } a_b = \sqrt{\frac{\beta_1}{1.7}} \cdot d \left\{ \begin{array}{l} f_c \leq 280 \text{ Ks/cm}^2 \\ \beta_1 = 0.85 \end{array} \right. \Rightarrow a_b = \frac{d}{\sqrt{2}} \quad (4)$$

Zona sísmica: $A_{smax} = 0.15 A_{sb}$
Debe cumplirse: $A_{smin} \geq A_{sl}$

Verificación por momento de agrietamiento M_{rt}
debe cumplirse: $1.5 M_{rt} < M_u (+)$

$$M_{rt} = f_r \cdot I_g$$

$$\left\{ \begin{array}{l} f_r = 2\sqrt{f_c} \text{ Esfuerzo de tracción concreto} \\ I_g = \frac{b \cdot h^3}{36} \text{ Momento de Inercia de la viga triangular} \end{array} \right.$$

$$\frac{M_u}{I} = \frac{M_u}{21.5}$$

PROBLEMA DE APLICACIÓN-1102

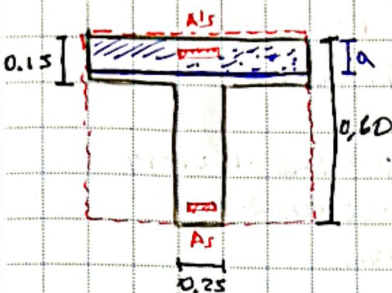
DISEÑAR LA VIGA TIPO T MOSTRADO, CONSIDERE:
RESPUESTA DEL ANÁLISIS ESTRUCTURAL $\rightarrow M_u = 24,50 \text{ ton}\cdot\text{m}$

$$f_c = 260 \text{ Kg/cm}^2 \quad f_y = 4200 \text{ Kg/cm}^2$$

NOTA: VUE N.T.P. ED60 RNE PARA CARGAS POR GRavedad

DATOS DE DISEÑO

$$d_1 = 54 \text{ cm} \quad d_2 = 51 \text{ cm} \quad K = 0,75 \quad \phi = 0,85$$



$$\text{Assume } d \leq h_f \Rightarrow b = 75 \text{ cm}$$

$$M_u \leq M_{ur \text{ max}} \rightarrow \text{D.S.R.}$$

$$M_u > M_{ur \text{ max}} \rightarrow \text{A.D.R.}$$

$$M_u = 24,50 \text{ ton}\cdot\text{m} < M_{ur \text{ max}} = 95,4855 \text{ ton}\cdot\text{m}$$

D.S.R.

*** CÁLCULO DE MÁXIMA CAPACIDAD DE RESISTENCIA**

$$P_b = 0,35 \beta_1 \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) = P_b = 0,02125; P_{\text{max}} = 0,75 P_b \Rightarrow P_{\text{max}} = 0,01594$$

$$A_{s \text{ max}} = P_{\text{max}} \cdot b \cdot d = 0,01594 \cdot 75 \cdot 51 \Rightarrow A_{s \text{ max}} = 60,961 \text{ cm}^2$$

$$A_{\text{min}} = \frac{A_{s \text{ max}} \cdot f_y}{0,85 \cdot f_c \cdot b} \Rightarrow A_{\text{min}} = 19,125 \text{ cm}^2$$

$$M_{ur \text{ max}} = \phi A_{s \text{ max}} \cdot f_y \left(d - \frac{A_{s \text{ max}} \cdot f_y}{2} \right) = 95,4855 \text{ Ton}\cdot\text{m}$$

*** CÁLCULO DEL ACERO DE DISEÑO**

$$\text{C.R.E.} = 0,006645$$

$$\text{C.R.E.} = 0,006645$$

$$0,59 w^2 - w + \frac{M_u}{\phi b d^2 f_c} = 0$$

$$w = 0,06928$$

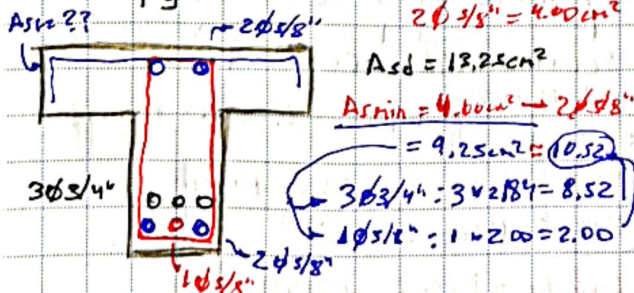
$$\text{② } \rho_b = w \frac{f_c}{f_y} = 0,006928 \frac{210}{4200} \Rightarrow \rho_b = 0,00346$$

$$\text{③ } A_s b = \rho_b \cdot b \cdot d = 0,00346 \cdot 75 \cdot 51 \Rightarrow A_s b = 13,25 \text{ cm}^2$$

$$\text{④ } A_d = \frac{A_s b \cdot f_y}{0,85 \cdot f_c \cdot b} = \frac{13,25 \cdot (4200)}{0,85 \cdot 210 \cdot 75} \Rightarrow A_d = 4,157 \text{ cm}^2 < \phi f$$

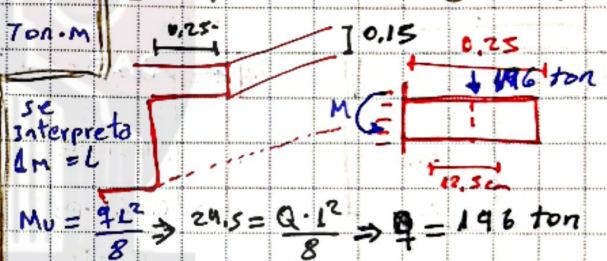
*** CÁLCULO DE ACERO MINIMO**

$$\rho_{\text{min}} = 0,00242; A_{s \text{ min}} = \rho_{\text{min}} \cdot b \cdot d = 3,26 \text{ cm}^2$$



TARGA CALCULAR EL ACERO EN ALA

CÁLCULO DE ACERO EN ALA DE VIGA T



EL MOMENTO QUE SE PRODUCE EN EL EMPOTRAMIENTO ES:

$$\Sigma M = 0 = M - 0,125 \text{ m} \times 196 \text{ ton}$$

$$M = 24,5 \text{ ton}\cdot\text{m}$$

SE TIENE QUE: $M_u = 24,5 \text{ ton}\cdot\text{m}$

$$M_u = K \cdot b \cdot d^2 \Rightarrow \text{Se interpreta } b = 1 \text{ m}$$

$$d = h - r = 0,15 - 0,06 \text{ m} = 0,09 \text{ m}$$

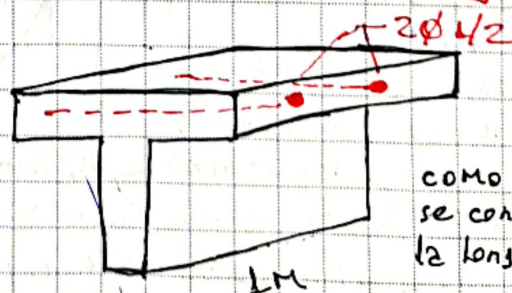
$$K = \frac{M_u}{b \cdot d^2} = \frac{24,5 \text{ ton}\cdot\text{m}}{(1 \text{ m} \cdot 0,09 \text{ m})^2} = 3024,69 \text{ ton/m}^2$$

Área de Acero $A_s = \rho \cdot b \cdot d$

$$A_s = 0,00242 (100 \text{ cm}) \cdot (9 \text{ cm})$$

$$A_s = 2,178 \text{ cm}^2 \Rightarrow 2\phi 1/2" = 2,58 \text{ cm}^2$$

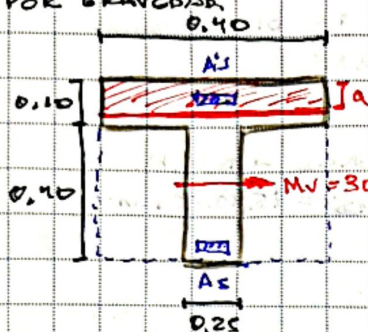
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COMO NO SE CONOCE LA LONGITUD.

PROBLEMA APLICATIVO #03:

DISEÑAR LA VIGA TIPO "T" CONSIDERE $f'_c = 210 \text{ Kg/cm}^2$, $f_y = 4200 \text{ Kg/cm}^2$, PARA UNA SOLICITUD ESTRUCTURAL $M_u = 30,50 \text{ ton}\cdot\text{m}$, NOTA: NTP 8.060 RNE, PARA CARGAS POR GRAVEDAD.



$d_1 = 44 \text{ cm}$ $d_2 = 41 \text{ cm}$ $K = 0,75$ $\beta_1 = 0,85$

ASUMO $a \leq h_f \rightarrow b = 40 \text{ cm}$

*** CÁLCULO CAPACIDAD MÁX RESISTENCIA**

$P_b = 0,85 \cdot \beta_1 \cdot \frac{f'_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow P_b = 0,02125$

$P_{máx} = 0,75 P_b = 0,75 (0,02125) \Rightarrow P_{máx} = 0,01594$

$A_{smáx} = P_{máx} \cdot b \cdot d = 0,01594 \cdot 40 \cdot 41 = 26,137 \text{ cm}^2$

$A_{máx} = \frac{A_{smáx} \cdot f_y}{0,85 \cdot f'_c \cdot b} \Rightarrow A_{máx} = 15,375 \text{ cm}$

{D.S.R.}

$M_u = 30,50 \text{ ton}\cdot\text{m} < M_{uRmáx} = 32,9127 \text{ ton}\cdot\text{m}$ $M_{uRmáx} = \phi A_{máx} \cdot f_y \left(d - \frac{a_{máx}}{2} \right) = 32,9127 \text{ ton}\cdot\text{m}$

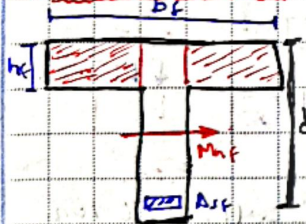
① $0,59w^2 \cdot w + \frac{M_u}{\phi b d^2 f'_c} = 0 \Rightarrow d/c = \frac{30,50 \cdot 10^5}{0,40 \cdot 40^2 \cdot 210} = 0,2400$ $\phi w = 0,28942$

② $\rho_b = \frac{w \cdot f'_c}{f_y} = 0,01447$; ③ $A_s d = \rho_b \cdot b \cdot d = 0,01447 \cdot 40 \cdot 41 \Rightarrow A_s d = 23,732 \text{ cm}^2$

④ $a_d = \frac{A_s d \cdot f_y}{0,85 \cdot f'_c \cdot b} = \frac{23,732 \cdot 4200}{0,85 \cdot 210 \cdot 40} \Rightarrow a_d = 13,96 \text{ cm} > h_f = 10 \text{ cm}$ **CASO III** $\rightarrow$ NO FUNCIÓN OTRA MÁS CASO 3

$M_uR = \phi A_{sf} \cdot f_y \left(d - \frac{h_f}{2} \right) + \phi A_{sw} \cdot f_y \left(d - \frac{a}{2} \right)$

*** ANÁLISIS EN ALA**



$C_f = 0,85 \cdot f'_c \cdot h_f (b_f - b_w)$

$J_d f = b \cdot \frac{h_f}{2}$

$T_f = A_{sf} \cdot f_y$

$C_f = T_f$

$0,85 \cdot f'_c \cdot h_f (b_f - b_w) = A_{sf} \cdot f_y$

$A_{sf} = \frac{0,85 \cdot f'_c \cdot h_f (b_f - b_w)}{f_y} = \frac{0,85 \cdot 210 \cdot 10 (40 - 25)}{4200}$

$M_{nF} = T_f \cdot J_d f$

$M_{nF} = A_{sf} \cdot f_y \left(d - \frac{h_f}{2} \right)$

$M_{nF} = 6,3775 \cdot 4200 \left(41 - \frac{10}{2} \right)$

$M_{nF} = 9,639 \text{ ton}\cdot\text{m}$

$M_{uRf} = \phi M_{nF} = 8,6751 \text{ ton}\cdot\text{m}$

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$M_{uRf} = \phi M_{nF} = 8,6751 \text{ ton}\cdot\text{m}$

$M_u = M_{uRf} + M_{uRw}$

$30,50 = 8,675 + M_{uRw}$

$M_{uRw} = 21,8241 \text{ ton}\cdot\text{m}$

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*** ANÁLISIS EN ALA**

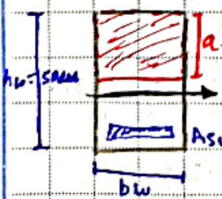
Datos Diseño:

$M_u = 21,8241 \text{ ton}\cdot\text{m}$

$b = b_w = 25 \text{ cm}$

$d_1 = 44 \text{ cm}$

$d_2 = 41 \text{ cm}$



$P_b = 0,85 \cdot \beta_1 \cdot \frac{f'_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow P_b = 0,02121$

$P_{máx} = 0,75 \cdot P_b \Rightarrow P_{máx} = 0,01594$

$A_{smáx} = P_{máx} \cdot b \cdot d = 0,01594 \cdot 25 \cdot 41 \Rightarrow A_{smáx} = 16,336 \text{ cm}^2 = A_{sw1}$

$A_{máx} = \frac{A_{smáx} \cdot f_y}{0,85 \cdot f'_c \cdot b} \Rightarrow A_{máx} = 15,375 \text{ cm}, C_{máx} = 18,888 \text{ cm}$

$M_{uRmáx} = \phi A_{smáx} \Rightarrow M_{uRmáx} = 20,57042 \text{ ton}\cdot\text{m}$

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$M_{uRmáx} = \phi A_{smáx} \Rightarrow M_{uRmáx} = 20,57042 \text{ ton}\cdot\text{m}$

$M_u \leq M_{uRmáx} \rightarrow \text{D.S.R.} \quad M_u > M_{uRmáx} \rightarrow \text{D.D.R.}$

$M_uR = \phi A_{s1} \cdot f_y \left(d - \frac{a}{2} \right) + \phi A_{s2} \cdot f_y \left(d - d' \right)$

SECTION REAL SECTION FICTICIA

$M_{uR} = M_{uR1} + M_{uR2}$

$21,8241 = 20,57042 + M_{uR2}$

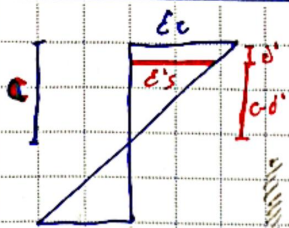
$M_{uR2} = 1,25448 \text{ ton}\cdot\text{m}$

ASUMIR: A_s fluye $\Rightarrow f_s = f_y = 4200 \text{ Kg/cm}^2$

$M_{uR2} = \phi A_{s2} \cdot f_y \left(d - d' \right)$

$A_{s2} = A_{s2} = \frac{M_{uR2}}{\phi f_y (d - d')} = \frac{1,25448 \cdot 10^5}{0,40 \cdot 4200 (41 - 6)}$

$A_{s2} = 0,94821 \text{ cm}^2 = A_{sw2}$



$$\frac{\epsilon_c}{c} = \frac{\epsilon_s}{c-d} \Rightarrow \epsilon_s = \frac{\epsilon_c(c-d)}{c} = \frac{0.003(18.888-6)}{18.888} \Rightarrow \epsilon_s = 0.0020$$

$$\epsilon_y = \frac{f_y}{E_s} = \frac{4200}{2 \times 10^6} \Rightarrow \epsilon_y = 0.0021 \quad \text{Since } \epsilon_s = 0.0020 < \epsilon_y = 0.0021$$

$$f_s = \epsilon_s \cdot E_s \Rightarrow f_s = 0.0020 \cdot 2 \times 10^6 \text{ Kg/cm}^2 \Rightarrow f_s = 4009.7561 \text{ Kg/cm}^2$$

$$A_s w 2 \cdot f_y = A'_s \cdot f_s$$

$$A'_s = A_s w 2 \cdot \frac{f_y}{f_s}$$

$$A'_s = 0.94821 \cdot \frac{4200}{4009.7561} \Rightarrow A'_s = 0.993 \text{ cm}^2$$

RESUMEN:

$$A'_s = 0.993 \text{ cm}^2$$

$$A_{sw} = A_{sw1} + A_{sw2} = 16.336 + 0.9482$$

$$A_{sw} = 17.284 \text{ cm}^2$$

$$A_s = A_{sf} + A_{sw} = 6.375 + 17.28$$

$$A_s = 23.659 \text{ cm}^2$$

* CALCULO DE ACERO MINIMO

$$P_{min} = \frac{0.90 \sqrt{f_c}}{f_y} \Rightarrow P_{min} = 0.00242$$

$$A_{smin} = P_{min} \cdot b \cdot d = 0.00242 \cdot 25 \cdot 44$$

$$A_{smin} = 2.657 \text{ cm}^2 \quad \text{2 } \phi 5/8" = 4.00 \text{ cm}^2$$

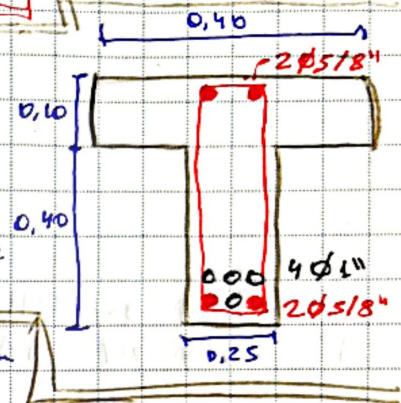
$$\text{Si } A'_s = 0.993 \text{ cm}^2 \quad A_{smin} = 2.657 \text{ cm}^2 \Rightarrow A_s = 4.00 \text{ cm}^2$$

$$A_s = 23.659 \text{ cm}^2$$

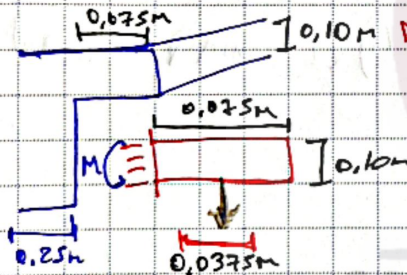
$$A_{smin} = 4.00 \text{ cm}^2$$

$$A_{SR} = 19.659 \text{ cm}^2$$

$$4 \phi 1" = 4(5.10) = 20.40 \text{ cm}^2$$



CALCULO DE ACERO EN ALA se interpreta $L = l_n$



$$M_u = 30.50 \text{ ton}\cdot\text{m}$$

$$M_u = \frac{qL^2}{8} \Rightarrow 30.50 = \frac{q \cdot 1^2}{8} = 244 \text{ ton} = q$$

El momento que se produce en el empotramiento es:

$$\sum M = 0 = M - 0.0375m \cdot 244 \text{ ton}$$

$$M = 9.15 \text{ ton}\cdot\text{m} \rightarrow \text{Se tiene que: } M_u = 9.15 \text{ ton}\cdot\text{m}$$

$$M_u = K \cdot b \cdot d^2 \rightarrow \text{se interpreta } b = l_n = L$$

$$d = h - r = 0.10 - 0.06 \text{ m} = 0.04 \text{ m}$$

$$K = \frac{M_u}{b \cdot d^2} = \frac{30.50 \text{ ton}\cdot\text{m}}{(1 \text{ m} \cdot 0.04^2)} = 19062.5 \text{ ton/m}^2$$

Por ello se puede decir que la cantidad resultará de igual forma con un valor bajo.

$$\text{Área de Acero } A_s = \rho \cdot b \cdot d$$

$$A_{sf} = 0.00242 \cdot 1 \text{ m} \cdot 0.04 \text{ m} \Rightarrow A_{sf} = 0.968 \text{ cm}^2$$

$$1 \phi 1/2" = 1.29 \text{ cm}^2 \rightarrow \text{Satisface}$$

